

INTEGRALES

— YisusReveck —

$$\int dx = x + C$$

$$\int k dx = kx + C$$

$$\int (u \pm v) dx = \int u dx \pm \int v dx$$

$$\int k f(x) dx = k \int f(x) dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C; \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C; \quad a > 0, a \neq 1$$

$$\int \ln x dx = x \ln x - x + C$$

$$\int \operatorname{sen} x dx = -\cos x + C$$

$$\int \cos x dx = \operatorname{sen} x + C$$

$$\int \tan x dx = -\ln |\cos x| + C$$

$$\int \cot x dx = \ln |\operatorname{sen} x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \csc x dx = \ln |\csc x - \cot x| + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \operatorname{sen}^2 x dx = \frac{1}{2}x - \frac{1}{4}\operatorname{sen}(2x) + C$$

$$\int \cos^2 x dx = \frac{1}{2}x + \frac{1}{4}\operatorname{sen}(2x) + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \operatorname{arcsen}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln |x + \sqrt{x^2 - a^2}| + C$$

$$\int \frac{x}{a^2 + x^2} dx = \frac{1}{2} \ln(a^2 + x^2) + C$$

$$\int u dv = uv - \int v du$$

L = Logarítmicas

I = Inversas

A = Algebraicas

T = Trigonómicas

E = Exponenciales

$$\int_a^b f(x) dx = F(b) - F(a)$$